📊 Business Recommendations — Customer Segmentation & Prediction

1. Executive Summary

Customer segmentation analysis identified three distinct groups: **Young Professionals**, **Budget Conscious**, and **Premium Spenders**. Predictive models achieved strong performance across all segments, with hyperparameter tuning further improving accuracy and F1-scores. These insights enable targeted business strategies to reduce churn and maximize customer lifetime value.

---

2. Segment-Specific Recommendations

Segment 1: Young Professionals

- **Profile:** Moderate tenure (32 months), high monthly charges ($127), but low total charges (~$1,870).
- **Model Performance:** Accuracy 94%, F1-score 0.77, ROC-AUC 0.85.
- **Business Strategy:**
  - Introduce **digital-first engagement** (mobile apps, self-service portals).
  - Offer **onboarding support** and loyalty incentives to increase retention.
  - Promote **bundled services** tailored to lifestyle needs (streaming, internet, mobile).
- **Expected Impact:** Improved retention among newer customers, reducing early churn risk.

Segment 2: Budget Conscious

- **Profile:** Long tenure (42 months), low monthly charges ($60), steady total spend (~$5,125).
- **Model Performance:** Accuracy 98%, F1-score 0.67, ROC-AUC 0.75.
- **Business Strategy:**
  - Provide **flexible payment plans** and targeted discounts.
  - Introduce **cost-saving bundles** to increase perceived value.
  - Focus on **customer service quality** to maintain loyalty.

- **Expected Impact:** Strengthened loyalty among price-sensitive customers, stabilizing revenue streams.

Segment 3: Premium Spenders

- **Profile:** Mid-tenure (36 months), high monthly charges ($156), high total spend (~$5,762).
- **Model Performance:** Accuracy 98%, F1-score 0.93, ROC-AUC 0.99.
- **Business Strategy:**
  - Launch **VIP loyalty programs** with exclusive offers.
  - Provide **premium support services** (priority helplines, dedicated managers).
  - Upsell and cross-sell **additional premium services**.
- **Expected Impact:** Increased revenue from high-value customers, enhanced brand loyalty.

---

3. Overall Business Impact

- **Retention Increase:** Targeted strategies expected to reduce churn across all segments.
- **Revenue Growth:** Premium segment strategies likely to drive upselling and cross-selling.
- **Customer Satisfaction:** Personalized engagement improves trust and long-term loyalty.

---

4. Conclusion

By tailoring strategies to each segment, the business can **maximize customer lifetime value**, **reduce churn**, and **strengthen competitive advantage**. The combination of clustering, predictive modeling, and hyperparameter tuning provides a robust foundation for actionable business insights.